

Ir⁺ ionisation limit 72323.90 cm⁻¹ = 8.96702 eV (138.267 nm)

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Ryd

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one e



Transition

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J er J' \rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$a^4F_{9/2} - 4038_{9/2}^o$	247.5871	247.5123	1210.8566	1.236	2.040e+07	-	-	first-step line of the in-gas-cell laser resonance ioniza...
$a^4F_{9/2} - z^6F_{11/2}^o$	351.4652	351.3647	852.9791	0.847	2.790e+06	0.4974	89.2	351 nm line

$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)
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Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$a\ ^4F_{9/2}$	62.655370(20)	426.23550(20)	F=3: -321.55	F=4: -355.08	F=5: -130.61	F=6: +529.48
$z\ ^6F_{11/2}$	308(4)	-750(30)	F=4: -3313.23	F=5: -1295.95	F=6: +674.77	F=7: +2353.50

Transition

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$a^4F_{9/2} - 4038^0_{9/2}$	247.587	193-191	-8100.00
$a^4F_{9/2} - z^6F^0_{11/2}$	351.465	193-191	-2000(10)